

VistaLux Entertainment Growth

Cooper Riggs, Adam Bitner, Stockton Nelson

Agenda

1. Introduction
2. Linear Model
3. Other Models
4. Model Performance
5. Understanding Growth
6. Conclusion

Meet the team



Dr. Matthew Heaton
Presenter



Cooper Riggs
Lead Data Analyst



Adam Bitner
Lead Programmer



Stockton Nelson
Assistant Data Analyst

Objective

This presentation forecasts VistaLux Entertainment's growth in 2025 and examines how key economic indicators — Unemployment, Savings, Production, Income, and Past Growth — affect performance. It compares multiple modeling approaches to provide the board with clear, actionable insights.

Previous Model

Linear Model

Summed the effects of income, savings, unemployment, and production.

$$y(t) = \beta_0 + \beta_1 \text{income}(t) + \beta_2 \text{savings}(t) + \beta_3 \text{unemployment}(t) + \beta_4 \text{production}(t) + \varepsilon_t$$

$y(t)$: company growth

β_1 : effect of income

β_2 : effect of savings

β_3 : effect of unemployment

β_4 : effect of production

ε_t : random error

No Seasonality

Didn't take into account how things could change or fluctuate over the year. Some quarters might just have higher growth regardless of those predictors (people spend more around Christmas for example)

Couldn't Project Future Data

Required information that was already known. We don't know what next years income, savings, unemployment, and production will look like.

Linear Model Predictions

By using a forecasting model to estimate future income, savings, production, and unemployment, the Linear Model Predicts this growth:

First Quarter	Second Quarter	Third Quarter	Fourth Quarter
0.3871594%	1.1884518%	0.6924379%	1.0970075%

Model Comparison

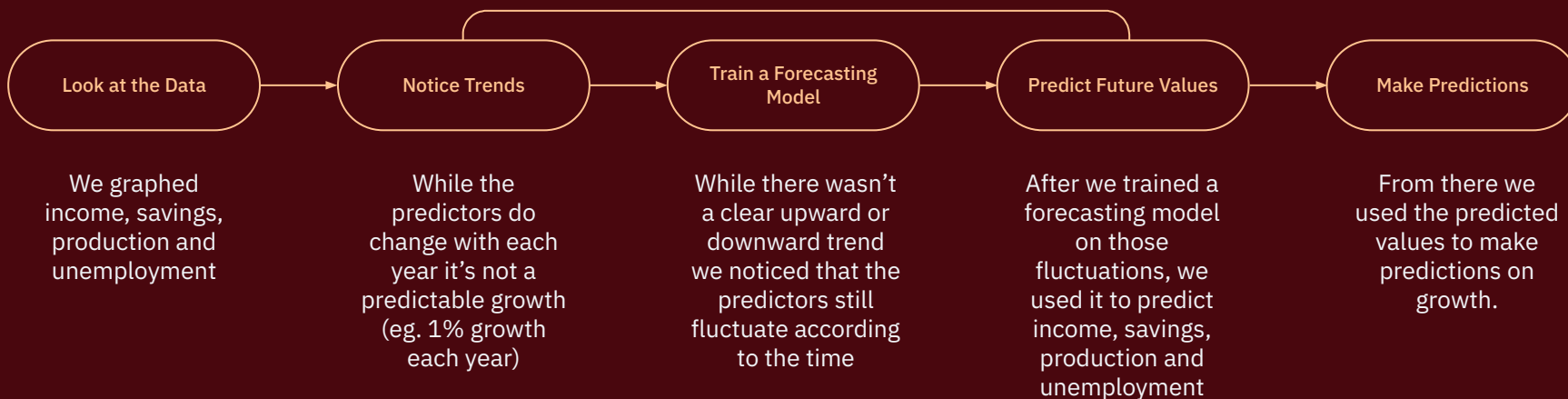
Facebook Prophet

SARIMAX

Model	Differences	Model	Differences
$y(t) = g(t) + s(t) + X\beta + \epsilon_t$ $y(t)$: Response variable (growth at time t) $g(t)$: Trend component (overall long-term growth, allows for changepoints) $s(t)$: Seasonal component (repeating patterns like quarterly effects) $X\beta$: Effects of external regressors (income, production, savings, unemployment) ϵ_t : Random error term	Unlike the linear regression model, which assumes a constant linear relationship and independent observations, the Prophet model decomposes the time series into trend and seasonal components and allows for non-linear growth and changepoints, making it better suited for data that evolves over time.	$y_t = \phi y_{t-1} + \theta \epsilon_{t-1} + X_t \beta + \epsilon_t$ $y(t)$: Response variable at time t (growth) ϕy_{t-1} : Autoregressive (AR) term — effect of the previous value of y $\theta \epsilon_{t-1}$: Moving average (MA) term — effect of the previous error $X\beta$: Effects of external regressors (income, production, savings, unemployment) ϵ_t : Random error term	In contrast to linear regression, which ignores temporal dependence, the SARIMAX model explicitly incorporates autocorrelation, seasonality, and past forecast errors while also allowing for external predictors, making it appropriate for time series data where observations are not independent.

How We Obtained Our Predictions

If you repeated steps or iterated on the process, change the conclusion step to show what you repeated.



Model Comparison

Model Evaluation Metric (RMSE)

* Lower = Better at Prediction

Linear Model

0.1588

Facebook Prophet

0.1680

SARIMAX

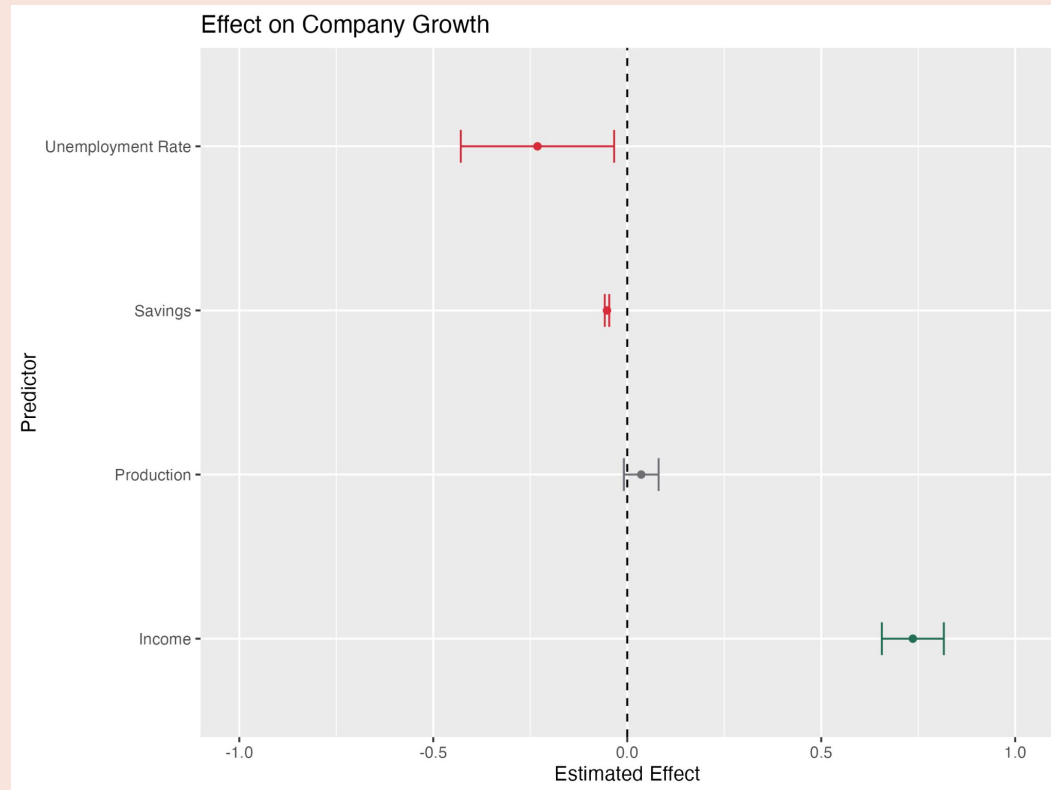
0.1374

Understanding Growth

Key Predictors in Explaining Growth with SARIMAX

For every unit increase, the expected growth is between...

- Unemployment Rate:
 - $(-0.429, -0.034)$
- Savings:
 - $(-0.058, -0.046)$
- Production:
 - $(-0.009, 0.081)$
- Income:
 - $(0.656, 0.816)$



Conclusions

****Recommended Model****

MODEL SUMMARY	Linear Model	Facebook Prophet Model	SARIMAX Model
SIMPLEST MODEL		-	
INCLUDES TIME ELEMENTS			
BEST FOR PREDICTION		-	

PREDICTOR SUMMARY	Income	Production	Savings	Unemployment
POSITIVE EFFECT				
NEUTRAL EFFECT				
NEGATIVE EFFECT				

Any questions?
Ask away!



Thank you!

Feel free to reach out if you have any further questions on our analysis.

Dr. Matthew Heaton
mheaton@stat.byu.edu